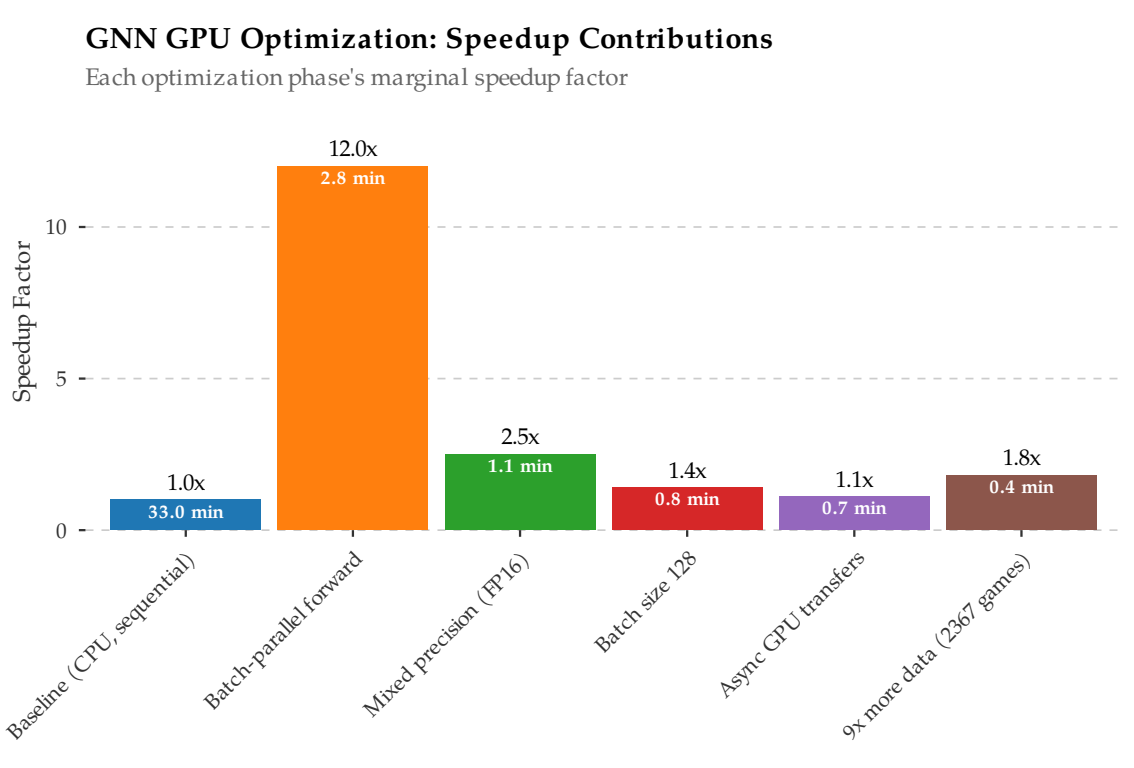
GNN GPU Optimization: Speedup Contributions
Each optimization phase's marginal speedup factor
Speedup Factor
0
5
10
1.0x
33.0 min
12.0x
2.8 min
2.5x
1.1 min
1.4x
0.8 min
1.1x
0.7 min
1.8x
0.4 min
Baseline (CPU, sequential)
Batch-parallel forward
Mixed precision (FP16)
Batch size 128
Async GPU transfers
9x more data (2367 games)